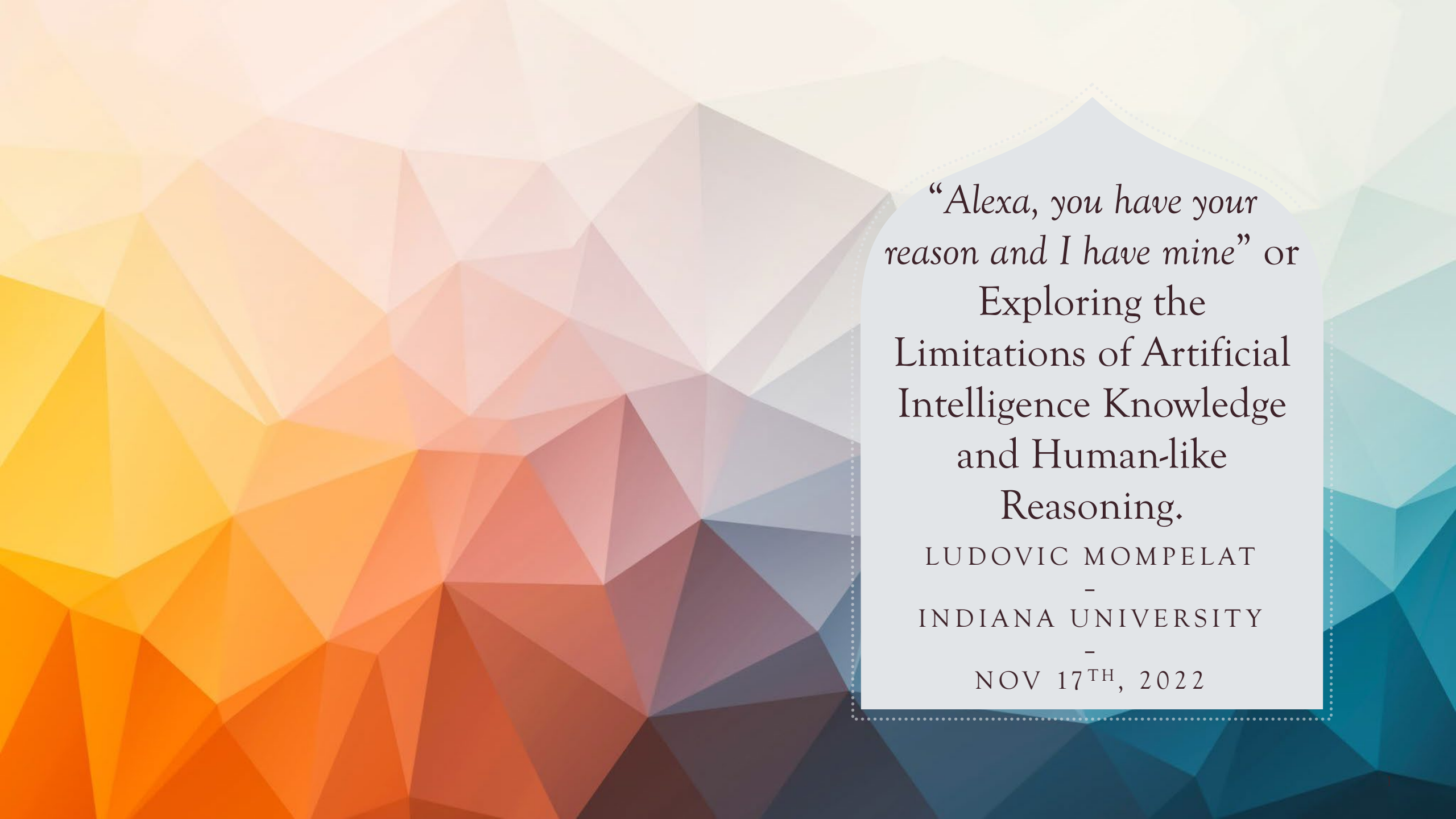




*“Alexa, you have your
reason and I have mine” or
Exploring the
Limitations of Artificial
Intelligence Knowledge
and Human-like
Reasoning.*

LUDOVIC MOMPELAT
–
INDIANA UNIVERSITY

–
NOV 17TH, 2022

“You have your reason and I have mine”

♦ **Artificial Intelligence Knowledge**

AI systems know only as much as the data we provide them with

- It cannot know what it hasn't been taught → Motivates developing NLP/CL research in Creoles

An AI model is only as good as its inner architecture and mostly the quality of the data

- It cannot make good data from bad data except if assisted → Motivates improving pre-existing datasets and providing deeper linguistic analysis

♦ **Human-like reasoning**

An AI model does not “reason” the same way humans do (think of parrots)

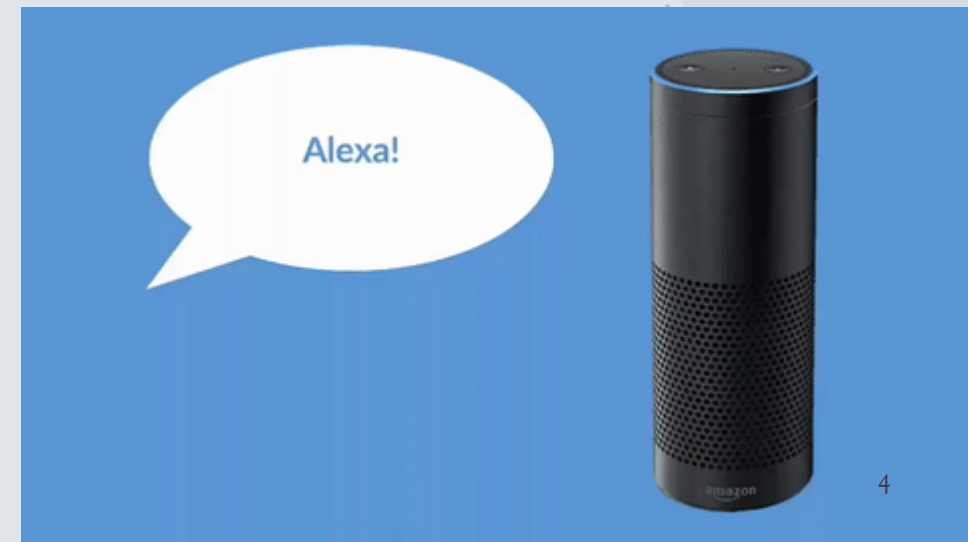
- It creates and establishes meaning of words based off of their linguistic context only → Motivates providing deeper semantic and reasoning depth to datasets

Overview

- ♦ What is Computational Linguistics?
- ♦ Research in Computational Linguistics
- ♦ Current Research
- ♦ Future directions

Computational Linguistics – Definition

- ♦ The use of computer science technology to the analysis, comprehension and generation of written and spoken language.
- ♦ At the intersection of linguistics, computer science, artificial intelligence, math and logic, cognitive science/psychology, etc.
- ♦ Extremely versatile field of research with an emphasis on interdisciplinary research



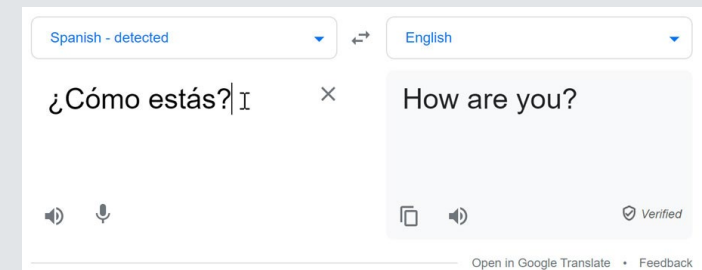
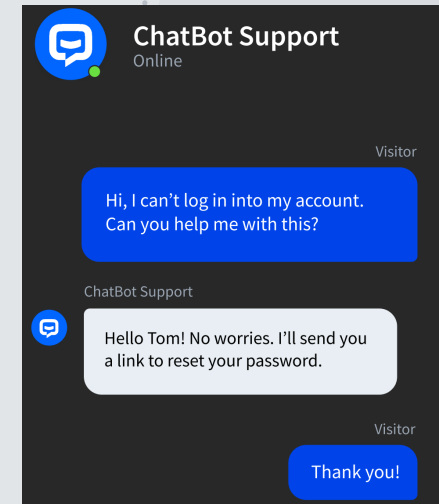
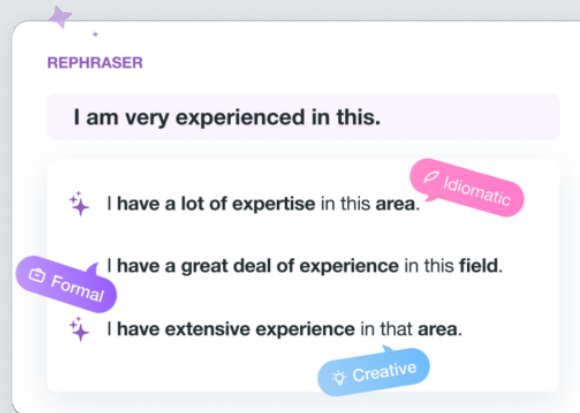
Putting the linguistics in “Computational linguistics”

Computational Linguistics’ contributions to the creation of practical tools :

- Grammar/Spell checker, Word prediction
- Automatic translators
- Language detection
- Sentiment analysis
- Chatbox
- Speech recognition
- Document indexing
- Text summarization/generation
- Anything Language related

Approaches to CL:

- Rule-based
- Statistical
- **Neural**



Putting the linguistics in “Computational linguistics”

Rule-based	Statistical	Neural/Deep learning
Explicit set of linguistics rules Human-made/readable	Statistical model (Decision trees, Naïve Bayes, SVM, etc.)	Neural Network
Machines apply rules in a systematic way	Supervised learning : Feature extractions by humans, learned by machines	Semi-supervised or unsupervised learning : machines learn features and rules
No learning process or word meaning representation	No learning process or word meaning representation	Word meaning representation and Transferability
Little to no data needed	Less data needed	Large amount of data and computational resources needed

How to Parse a Creole: When Martinican Creole Meets French

Ludovic Mompelat, Daniel Dakota, and Sandra Kübler. 2022.

Basics of syntactic parsing : automatic analysis of the syntactic structure of languages

- Part of Speech tagging and dependency parsing

“Sé yo ka fè’w bonjou” (*They are the ones greeting him*)

- Provides syntactic structure of a sentence and deeper linguistic understanding

- Useful for tasks such as :

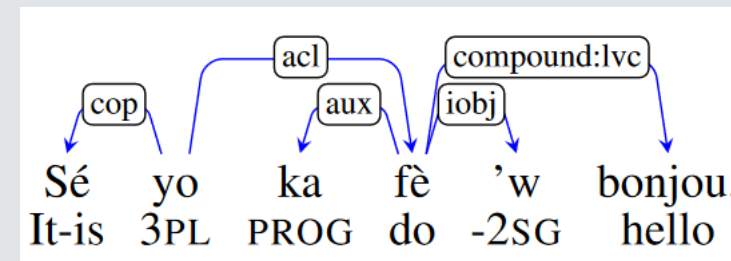
Machine translation

Word prediction

Grammar/Spellchecker

Etc.

- Long and tedious task



Token	POS
Sé	VERB
yo	PRON
ka	PART
fè	VERB
,	PUNCT
w	PRON
bonjou	NOUN
.	PUNCT

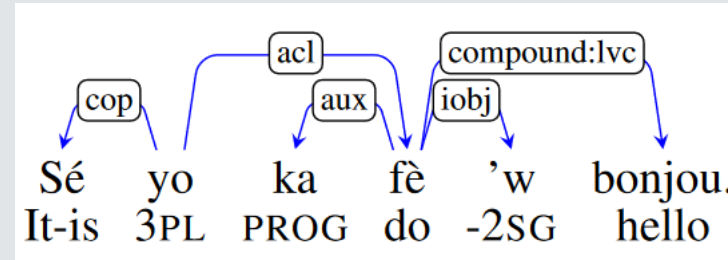
Universal Dependency Treebank for Martinican Creole

Ludovic Mompelat and Sandra Kübler. Peer Reviewed.

- Developing a parser splits into 2 tasks :

Data collection and annotation → Treebank generation

Parser creation → Using neural approaches



Token	POS
Sé	VERB
yo	PRON
ka	PART
fè	VERB
,	PUNCT
w	PRON
bonjou	NOUN
.	PUNCT

Treebank efforts in the community :

Universal Dependencies (Nivre et al. 2020) → Taxonomy of grammatical relations applicable to a variety of languages.

Complex, costly and time-consuming annotation task.

→ Not an issue for dominant languages : 45k English sentences, 48k French sentences

→ What of under-resourced and under-represented languages, like creole languages?

240 MC sentences annotated for POS and dependencies



How to Parse a Creole: When Martinican Creole Meets French

Ludovic Mompelat, Daniel Dakota, and Sandra Kübler. 2022.

- ♦ Syntactic parsing splits into 2 tasks :

Data collection and annotation → Treebank generation

Parser creation → Using neural approaches

Neural models demand large amount of data split into Train, Dev and Test sets.

240 sentences are not enough (8:1:1 div) 192:24:24

How to make the most of this extremely limited corpus?

Using a treebank of a language close enough to MC and which benefitted from greater annotation efforts → French

Train set = the lesson (80%)

Dev set = the practice quiz (10%)

Test set = the real quiz (10%)



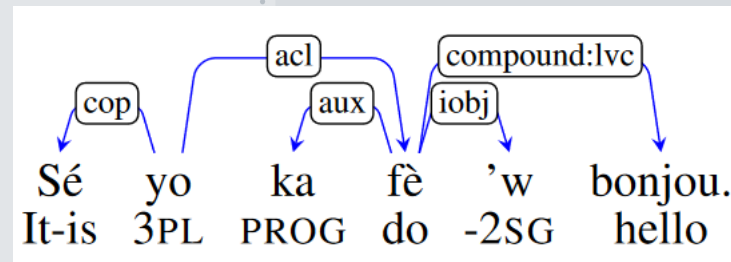
How to Parse a Creole: When Martinican Creole Meets French

Ludovic Mompelat, Daniel Dakota, and Sandra Kübler. 2022.

Experiment specs :

- 240 MC sentences annotated for POS and dependencies
News and blog articles from 2004 and 2021 (4809 tokens)
Redefine the split into 3 averaged randomized splits 3:3:3
- French corpus of 16k sentences to leverage (French GSD corpus)
- Character, POS and camemBERT embeddings (Martin et al. 2020)
- 2 approaches to leveraging French for parsing :
 - 1. Baseline
 - 2. Transfer learning
 - 3. Multi-task learning

Train	Dev	Test
French 12,8k	MC 80	MC 80
MC 80	MC 80	MC 80
French+MC 12,8k+80	MC 80	MC 80



Train	Embed.	UAS	LAS
French	char	25.05	11.73
	POS	65.08	51.95
	BERT	38.23	21.63
MC	char	62.89	48.36
	POS	71.71	62.86
	BERT	63.36	49.83
FR+MC	char	72.95	60.57
	POS	80.75	71.77
	BERT	72.17	58.57

Table 2: Baselines for training on French, Martinican (MC), and concatenated French+Martinican (FR+MC).

How to Parse a Creole: When Martinican Creole Meets French

Ludovic Mompelat, Daniel Dakota, and Sandra Kübler. 2022.

2 approaches to leveraging French for parsing :

- 1. Baseline
- 2. Transfer learning : Improving the French baseline
- 3. Multi-task learning

Train a first model, then retraining it for a different task

- Does not start the model from scratch, allows it to be more specialized

Optimization and overfitting issues :

- Using French dev set for baseline optimization vs MC
- French optimized model shows signs of overfitting

Train	Dev		Train	Dev	Test
Baseline		→	Fine-tuning/Retraining		
French	French		MC	MC	MC
French	MC				

MC baseline POS
UAS: 71,71 LAS:62,86
FR+MC baseline POS
UAS: 80,75 LAS:71,77

Dev.	Embed.	Finet.	UAS	LAS
French	char	no	20.03	9.16
		yes	20.03	9.16
	POS	no	58.17	45.54
		yes	58.17	45.54
	BERT	no	33.07	18.38
		yes	33.07	18.38
MC	char	no	25.05	11.73
		yes	64.71	46.83
	POS	no	65.08	51.95
		yes	72.87	60.83
	BERT	no	38.23	21.63
		yes	67.10	49.41

Table 4: Performance with and without fine-tuning on MC.

How to Parse a Creole: When Martinican Creole Meets French

Ludovic Mompelat, Daniel Dakota, and Sandra Kübler. 2022.

- 2 approaches to leveraging French for parsing :
 - 1. Baseline
 - 2. Transfer learning : Improving the French baseline
 - 3. **Multi-task learning**
- Train two models at the same time for joint optimization
- Combines information between two treebanks
- Loss weighting to reduce negative transfer : Great negative transfer for POS tasks

MC baseline POS
 UAS: **71,71** LAS:**62,86**
FR+MC baseline POS
 UAS: **80,75** LAS:**71,77**

Embeddings	Weights		UAS	LAS
	FR	MC		
char	0.90	0.10	71.44	58.26
POS	0.90	0.10	64.99	50.12
BERT	0.90	0.10	70.76	56.78
	FR+MC	MC		
char	0.90	0.10	73.13	59.84
POS	0.90	0.10	69.56	55.81
BERT	0.90	0.10	73.37	60.08

Table 7: Results for the MC task using varying weights for the MTL parser, training on either FR and MC or on FR+MC and MC and testing on MC.

MTL	Train	Dev	Test		Train	Dev	Test
	Task 1			↕	Task 2		
Experiment 1	French	MC	MC		MC	MC	MC
Experiment 2	French+MC	MC	MC		MC	MC	MC

How to Parse a Creole: When Martinican Creole Meets French

Ludovic Mompelat, Daniel Dakota, and Sandra Kübler. 2022.

Conclusions :

- Expected limitations from a limited corpus
- Best performances from Corpus concatenation & MTL

Surprisingly 240 sentences can already yield high results (63% - 72%)

Future directions :

- Implement phonetic representations for French and for Creole
- Annotate more sentences for Gold standard dataset



Gold standard vs Silver standard

- Previous research's goal → provide large amount of silver standard data at a much faster pace than sufficient gold standard data.
- Troubleshooting, checking and balancing the silver data > annotate gold data from scratch.
- Silver data with 95-98% accuracy is good enough for a lot of NLP tasks



How “Loco” is the LOCO Corpus? Annotating the Language of Conspiracy Theories

Ludovic Mompelat, Zuoyu Tian, Amanda Kessler, Matthew Luetzgen, Aaryana Rajanala, Sandra Kübler, and Michelle Seelig. 2022.♦

LOCO (Language Of Conspiracy) Corpus :

- ♦ 100k documents webscraped (88mil tokens)
- ♦ Set of seeds representing different Conspiracy Theories (CT)
- ♦ **Conspiracy subcorpus** = CT websites based on highest Conspiracy score.
- ♦ **Mainstream subcorpus** = Google search of the seeds and domain extraction.
- ♦ **Paper’s goal** → Evaluate the automatic retrieval methods used for the corpus
- ♦ **Ultimate goal** → Using the LOCO corpus to develop ML approaches to CT detection



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Ludovic Mompelat, Zuoyu Tian, Amanda Kessler, Matthew Luetngen, Aaryana Rajanala, Sandra Kübler, and Michelle Seelig. 2022.♦

2 main tasks :

- ♦ Creating an annotation schema to categorize documents as containing Conspiracy Theory content/language/belief or not.
- ♦ Compare our annotations to the automatic labelling of documents
- ♦ 2 rounds of annotation – 5 annotators – Sandy Hook seed

Mainstream subcorpus = High inter-annotator agreement → Automatic retrieval methods are efficient

Conspiracy subcorpus = Low inter-annotator

No clear distinction between reports of events, reports of CT related to the event, **propagation of CT**

→Need for conspiracy belief

Topic	Category	Round	5 ann. agree	4+ ann. agree
Sandy Hook	CT	1	9/20	17/20
	mainstream	1	18/20	20/20



How “Loco” is the LOCO Corpus? Annotating the Language of Conspiracy Theories

Ludovic Mompelat, Zuoyu Tian, Amanda Kessler, Matthew Luetzgen, Aaryana Rajanala, Sandra Kübler, and Michelle Seelig. 2022♦

Defining Conspiracy Theory content/language/belief or not.

- ♦ **Content** = The theme of the document, does it talk about a conspiracy?

Contradictory opinion to mainstream opinion / sensationalism / mention of other CTs

- ♦ **Language** = How is the document linguistically constructed?

Use of caps/atypical named entities / unconventional punctuation!!!/Frequent use of 1st and 2nd person pronouns / Frequent questions directed at the reader / paraphrasing instead of direct quoting

- ♦ **Belief** = Does the author of the document believe and seek to propagate the conspiracy?

- “FBI *says* No One Killed at Sandy Hook” [LOCO ID: C005a9]
- “They reported Fake numbers that they made up & don’t even exist. **WE WILL WIN AGAIN!**” [LOCO ID: C00775]
- “The Obama White House” [LOCO ID: C00a2d]
- “**‘Sleepy Joe’** makes another gaffe on his campaign trail” [LOCO ID: C0690d]
- “If you want more evidence of a **government seeking control**, look no further than the IRS scandal where the Obama administration was using the IRS to **stop conservatives and religious groups** from organizing opposition.” [LOCO ID: C00650]

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Ludovic Mompelat, Zuoyu Tian, Amanda Kessler, Matthew Luetzgen, Aaryana Rajanala, Sandra Kübler, and Michelle Seelig. 2022.♦

- ♦ Applying our definition and textual cues
Higher Inter-annotator agreement onto round 2 for Sandy Hook
- ♦ Testing robustness of Textual and context cues → Using Coronavirus conspiracy
Lower inter-annotator agreement than SH

Different CTs = different document structures

Topic	Category	Round	5 ann. agree	4+ ann. agree	Fleiss' kappa	Krippendorff's alpha
Sandy Hook	CT	1	9/20	17/20	0.466	0.469
	mainstream	1	18/20	20/20	-0.020	-0.010
Sandy Hook	CT	2	14/20	17/20	0.696	0.699
	mainstream	2	20/20	20/20	1.0	1.0
Coronavirus	CT		12/20	17/20	0.577	0.575
	mainstream		19/20	20/20	-0.010	0

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Ludovic Mompelat, Zuoyu Tian, Amanda Kessler, Matthew Luetzgen, Aaryana Rajanala, Sandra Kübler, and Michelle Seelig. 2022♦

2 main tasks :

- ♦ Creating an annotation schema to categorize documents as containing Conspiracy Theory content/language/belief or not.
- ♦ Comparing our annotations to the automatic labelling of documents

Seed	Category	# LOCO docs	# docs annotated	Conspiracy Rate
Sandy Hook	Conspiracy	364	364	0.615
Sandy Hook	Mainstream	1476	200	0.020
Coronavirus	Conspiracy	571	571	0.413
Coronavirus	Mainstream	1965	20	0

Table 4: Statistics of our annotations.

How “Loco” is the LOCO Corpus? Annotating the Language of Conspiracy Theories

Ludovic Mompelat, Zuoyu Tian, Amanda Kessler, Matthew Luetngen, Aaryana Rajanala, Sandra Kübler, and Michelle Seelig. 2022.♦

Conclusions :

- Re-evaluate the automatic method of data collection
- Annotation schema allows for more accuracy and agreement in document classification

Looking forward :

- Include concept of Conspiracy Thinking (Introne et al. 2020)
 - *“Documents that imply a conspiracist point of view [...] but do not themselves contain identifiable CTheories”*
- Develop ML solutions for CT belief detection, to detect CT propagation



Dangerous liaisons between a model and its data

- An NLP tool or AI model is only as good as the data it is given
- High quality data creation efforts vs. unsupervised approach with extreme quantity of data

→ ChatGPT, based on GPT-3.5 model architecture with 175 billion parameters

- Language models and AIs cannot be 100% accurate on a given task.
- Semantic reasoning is still the more obscure part of NLP.



Event Sequencing Annotation with TIE-ML

Damir Cavar, Ali Aljubailan, Ludovic Mompelat, Yuna Won, Billy Dickson, Matthew Fort, Andrew Davis, and Soyoung Kim. 2022.

- ♦ Event sequencing annotation :

Wash the veggies, 1	chop them, 2	and fry them 3
Before you fry the veggies, 3	wash them, 1	and chop them 2

- ♦ Some event sequencing processes are universal, and/or dictated by the syntactic structure of a given clause. These clues are detected by humans, but not so much by computer systems.

Event Sequencing Annotation with TIE-ML

Damir Cavar, Ali Aljubailan, Ludovic Mompelat, Yuna Won, Billy Dickson, Matthew Fort, Andrew Davis, and Soyoung Kim. 2022.

John **said** that he **left** early

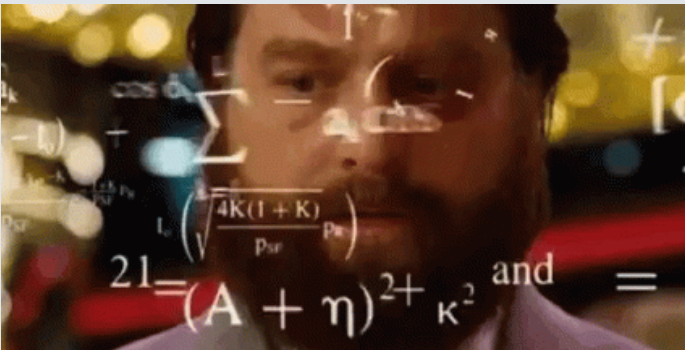
2 1

John **believes** Mary to **have left**

2 1

John is **watching** Mary **leave** the store

John **makes** Mary **eat** the fish



LM

In the sentence "John is watching Mary leave the store", which action happens first, see or leave?



In the sentence "John is watching Mary leave the store," the action of "leave" happens first.

LM

In the sentence "John makes Mary eat the fish", which action happens first, make or eat ?



In the sentence "John makes Mary eat the fish," the action of "make" happens first.

LM

In the sentence : John said that he left early, which action happens first, say or leave ?



It is not specified in the sentence which action happened first, whether John "said" or "left early." The sentence only states that John said he left early, it doesn't say when he said it or when he left.

Event Sequencing Annotation with TIE-ML

Damir Cavar, Ali Aljubailan, Ludovic Mompelat, Yuna Won, Billy Dickson, Matthew Fort, Andrew Davis, and Soyoung Kim. 2022.

- **Ultimate goal** : Annotate a large quantity of data with enough features that a model can make its own connections and links.
- Need for a systematic representation/schema for temporal information annotation.
- TIME-ML (Pustejovsky et al. 2003) exists but too costly to use

(11) John taught 20 minutes every Monday.

```
John
<EVENT eid="e1" class="OCCURRENCE">
taught
</EVENT>
<MAKEINSTANCE eiid="ei1" eventID="e1" pos="VERB" tense="PAST" aspect="NONE" polarity="POS"/>
<TIMEX3 tid="t1" type="DURATION" value="P20TM">
20 minutes
</TIMEX3>
<TIMEX3 tid="t2" type="SET" value="xxxx-wxx-1" quant="EVERY">
every Monday
</TIMEX3>
<TLINK timeID="t1" relatedToTime="t2" relType="IS_INCLUDED"/>
<TLINK eventInstanceID="ei1" relatedToTime="t1" relType="DURING"/>
```

Event Sequencing Annotation with TIE-ML

Damir Cavar, Ali Aljubailan, Ludovic Mompelat, Yuna Won, Billy Dickson, Matthew Fort, Andrew Davis, and Soyoung Kim. 2022.

A clause must have one and only one predicate. Each predicate is marked for tense and aspect.

- **Clause ID** : linear order of a given clause
- **Time Slot** : chronological order of a given clause
- **Speaker, Event times** : Reichenbach's theory of tense
- **Level** : Independent/main clause (=0), subordinate clause (=level of main clause -1)
- **Selected by** : Clause ID of the dominating clause

Annotation schema

Clause ID

Time Slot

Speaker time

Event time

Level

Selected by

9 The yield on six-month Treasury bills **Pass.** sold at Monday 's auction , for example , **Simp. | Past** rose to 8.04 % from 7.90 % .

(Clausal annotations)

TEMPpart

1 | 2 | 0 | -1 | 0 | 0

2 | 1 | 0 | -1 | 1 | 1

1 | 2 | 0 | -1 | 0 | 0

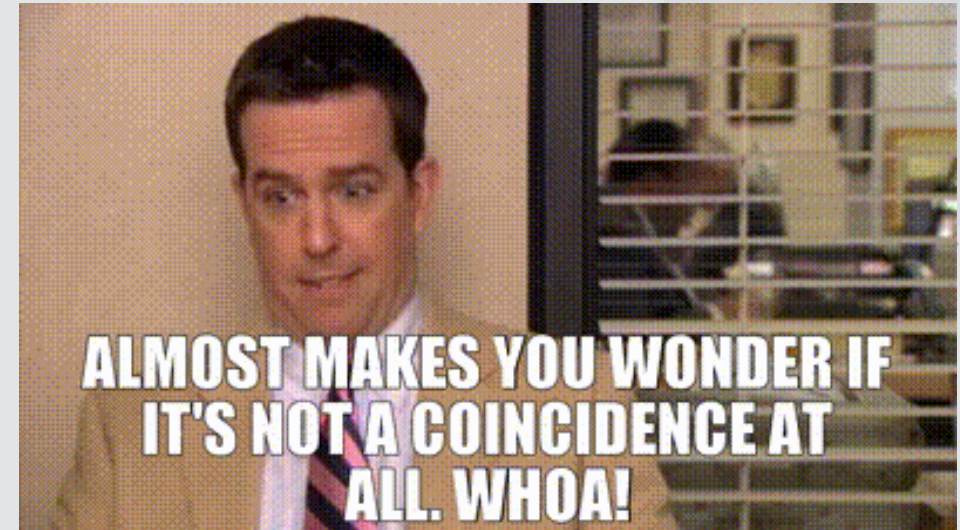
Event Sequencing Annotation with TIE-ML

Damir Cavar, Ali Aljubailan, Ludovic Mompelat, Yuna Won, Billy Dickson, Matthew Fort, Andrew Davis, and Soyoung Kim. 2022.

Conclusions : Simplified but not simplistic schema

- 1st step towards understanding relationship between the temporal specification of a main clause's predicate wrt that of the subordinate clause's predicate.
- It is rooted in the very notion of finiteness, and the description of the complementation system of natural languages
- It is the topic of my own dissertation :

To Infinitive and Beyond, or Revisiting Finiteness in Creoles



To Infinitive and Beyond, or Revisiting Finiteness in Creoles

Contrastive study of syntactic structure of complements in Martinican Creole and Haitian Creole.

Creation of the first annotated corpus, parser and POS tagger for these languages

- Claims that there is no distinction between finite and non-finite (infinitive) clauses
- Weigh in on the debate surrounding the linguistic concept of finiteness for creole languages, and relate it to other languages and language phenomena
- Looking at data in the literature from the past 40 years + data collected from newspapers, blogs, and native speakers

John **is sleeping**
John **slept**
John said he **slept**

John wants him to **sleep**
John makes him **sleep**

Jan **ka dòmi**
Jan **dòmi**
Jan di i **dòmi**

Jan lé i **dòmi**
Jan ka fè i **dòmi**

Looking at : temp. specification /subj. licensing

Axes	Approach	Methodology	Analysis & Discussion	Conclusion
Theoretical Linguistics	Finiteness as syntactic and semantic processes	Predicate choices : causatives, want-class, seem-class, modals	Negation, subject licensing, tense/aspect marking and scope Complement restructuring	- Do Creole languages really have no finite/nonfinite distinction?
Applied Creole Linguistics	Studying Martinican Creole, Haitian Creole and French structures	Different complement structures in the literature Consultants acceptability judgments	Diachronic comparison Sociolinguistic variation Language Contact	- What can creole languages contribute to the definition of finiteness ? - How does developing NLP tools for creole languages help develop further creole research, and can contribute to cross-linguistic comparisons and collaborations?
Computational Linguistics	Compare Martinican Creole with Haitian Creole and French	Annotated Corpus, POS tagger and parser	Synchronic comparison How corpus informs theories	

Looking further

- ♦ **Computational Linguistics**

How to best balance the corpus annotation efforts of an under-resourced language and leveraging available data from a related language to achieve SotA parsing scores?

- ♦ **Corpus Linguistics**

How can corpus generation and annotation efforts help safeguard and promote minority languages? How can we implement such tools in the classroom?

- ♦ **Creole Linguistics / Sociolinguistics / Language Contact**

How can the reanalysis of the creole continua inform us on the impact of the power dynamics between France/French and the Caribbean territories/languages in contact?

- ♦ **Theoretical Linguistics**

How can creole languages help challenge and push the boundaries of the current theories in syntax regarding finiteness?

Thank you! 😊

Some references to get started

- Textbook : Speech and Language Processing (3rd ed.) Dan Jurafsky and James H. Martin
<https://web.stanford.edu/~jurafsky/slp3/>
- Cool Tools
 - Wordnet : <http://wordnetweb.princeton.edu/perl/webwn>
 - Framenet : https://framenet.icsi.berkeley.edu/fndrupal/framenet_search
 - Stanford Corenlp : <https://corenlp.run/>
- Corpus Linguistics tools :
 - Concordancer <https://www.lextutor.ca/>
 - Wordsmith <https://wordsmith.org/>
 - Antconc <https://www.laurenceanthony.net/software/antconc/>
 - British National Corpus (BNC) <https://www.english-corpora.org/bnc/>
 - The Corpus of Contemporary American English (COCA) <https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/AMUDUW>